

Community Lost & Found Matching Network

FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match newly logged Found Item posts with active Lost Item reports based on category, colour tags, and location radius.	High	Pass: For a test Found Item with compatible category/colour and location within the configured radius, the system produces a match and records a match probability. Fail: an incompatible category is matched.	Core functionality for connecting lost and found reports.
FR-002	Functional	The system shall allow a Finder/Owner to create and submit a Lost Item or Found Item report containing item details, location, and available visual/text information.	High	Pass: A valid report is saved and displayed in the user's active reports. Fail: the system accepts a submission missing a required field.	Provides the reports required for matching.
FR-003	Functional	The system shall extract visual and text item tags from submitted item information and associate the extracted tags with the corresponding report.	Medium	Pass: For a test image/text input containing known category and colour information, the corresponding tags are stored with the report. Fail: tags are not associated with the correct report.	Tags are required as matching attributes in the scenario.
FR-004	Functional	The system shall notify a potential owner when a Lost Item report receives a qualifying match with a Found Item report.	Medium	Pass: After a qualifying match is generated, a notification is dispatched to the potential owner. Fail: no notification is generated for a qualifying match.	Allows potential owners to act on detected matches.
FR-005	Functional	The system shall require a claimant to answer the private security questions set by the Finder before revealing handover contact details.	High	Pass: Correct answers reveal the permitted contact details. Fail: incorrect answers leave the contact details hidden.	Prevents unauthorized claims and supports proof-of-ownership verification.
NFR-001	Nonfunctional	Claimants must answer private security questions set by the Finder before item handover contact details are revealed.	High	Pass: Testing confirms that contact details remain hidden until the required security-question verification succeeds under simulated peak load.	Protects the Finder's privacy and provides an ownership-verification control.
NFR-002	Nonfunctional	The system shall calculate a lost-found match and dispatch the corresponding notification within 5 seconds for 95% of matching requests under simulated peak load.	High	Pass: A performance test shows at least 95% of matching requests complete with notification dispatch within 5 seconds. Fail: the measured rate is below 95%.	Keeps the matching network responsive enough for practical community use.

USE CASE SCENARIO

Use Case: Claim Found Item

Primary Actor: Finder/Owner (claimant)

Supporting Actor: Community Admin

Purpose: Allow a potential owner to claim a found item after the system verifies proof of ownership using private security questions.

Preconditions

A Found Item report is active in the system.

The claimant has received or located a qualifying match.

The Finder has configured private security questions for the item.

Postconditions

If verification succeeds, the permitted handover contact details are revealed.

If verification fails, the handover contacts details remain hidden and the claim is not completed.

Main Success Scenario

1. The claimant opens the qualifying Found Item match.
2. The system displays the item information and starts the ownership-verification process.
3. The system presents the private security questions configured by the Finder.
4. The claimant enters answers to the security questions.
5. The system validates the submitted answers.
6. The system confirms successful ownership verification.
7. The system reveals the permitted handover contact details.
8. The claimant and Finder can proceed with item handover coordination.
9. The use case ends successfully.

Alternate Flow — Incorrect Security Answers

- 5a. The system determines that one or more answers are incorrect.
 - 5a1. The system displays an ownership-verification failure message.
 - 5a2. The handover contacts details remain hidden.
 - 5a3. The claimant may retry the verification according to the system's configured policy.

5a4. If verification does not succeed, the claim remains incomplete.

Relationship Used in the UML Model

Claim Found Item «include» Verify Ownership, because ownership verification is required before handover contact details are revealed.

UML DIAGRAM

